(6.4) holds. Taking the inner product of $x_i(\frac{1}{n})$ with the decomposition (6.4) at time $t = \frac{1}{n}$, we get

$$\alpha\left(\frac{1}{n}\right) \geqslant \min_{(i,j)\in[n]^2} \left\langle x_i\left(\frac{1}{n}\right), x_j\left(\frac{1}{n}\right)\right\rangle \geqslant \gamma_\beta\left(\frac{1}{n}\right) - 2c(\beta)\sqrt{\frac{\log(d)}{d}}$$
$$\geqslant \frac{1}{2}\gamma_\beta\left(\frac{1}{n}\right),$$

where the second inequality comes from (D.6) evaluated at time $t = \frac{1}{n}$, and the last inequality comes from (D.9). This is precisely (D.11). Using the notation $a_{ij}(t) = Z_{\beta,i}(t)^{-1}e^{\beta\langle x_i(t), x_j(t)\rangle}$ as in the proof of Lemma 6.4, we now find

$$\dot{\alpha}(t) = \langle \dot{x}_{i(t)}(t), x^*\rangle \geqslant \sum_{j=1}^{n} a_{i(t)j}(t)(1 - \langle x_{i(t)}(t), x_j(t)\rangle)\alpha(t) \tag{D.12}$$

for one of the indices $i(t) \in [n]$ achieving the minimum in the definition of $\alpha(t)$. Combining this with (D.11), we gather that $\alpha(t) \geqslant \alpha(\frac{1}{n})$ for $t \geqslant \frac{1}{n}$. But

$$\min_{j\in[n]} \langle x_{i(t)}(t), x_j(t)\rangle \leqslant \sum_{k=1}^{n} \theta_k(t)\langle x_{i(t)}(t), x_k(t)\rangle = \langle x_{i(t)}(t), x^*\rangle = \alpha(t). \tag{D.13}$$

Plugging (D.13) into (D.12) and using $a_{ij}(t) \geqslant n^{-1}e^{-2\beta}$ we get

$$\dot{\alpha}(t) \geqslant \frac{1}{ne^{2\beta}}\alpha\left(\frac{1}{n}\right)(1 - \alpha(t)) \tag{D.14}$$

for $t \geqslant \frac{1}{n}$. Integrating (D.14) from $\frac{1}{n}$ to t, we get (D.10). We therefore deduce from (D.10) that

$$\langle x_i(t), x_j(t)\rangle \geqslant 1 - \exp\left(\frac{1 - \gamma_\beta(\frac{1}{n})t}{2ne^{2\beta}}\right)$$

holds for any distinct $i, j \in [n]$. Together with (D.8), we then get

$$\left|\langle x_i(t), x_j(t)\rangle - \gamma_\beta(t)\right| \leqslant \exp\left(\frac{1 - \gamma_\beta(\frac{1}{n})t}{2ne^{2\beta}}\right) + \frac{1}{2}\exp\left(\frac{n^2e^\beta}{2(n + e^{\frac{\beta}{2}})} - \frac{nt}{n + e^{\frac{\beta}{2}}}\right). \tag{D.15}$$

Finally, combining (D.7) and (D.15) we obtain (6.11). □

Remark D.1. *An analogous statement to Theorem 6.9 holds for* (USA)*, where* γ_β *would rather be the unique solution to* (6.10)*. More concretely, Step 1 in the proof is only slightly changed—the constant one obtains in the analogue of* (6.11) *is rather* $c(\beta)^{nt}$ *with* $c(\beta) = e^{10\beta e^{2\beta}}$ *—. Step 2 remains unchanged. In Step 3,* (D.8) *is replaced by* $\gamma_\beta(\frac{n}{2}) \geqslant \frac{1}{2}$ *and*

$$1 - \gamma_\beta(t) \leqslant \frac{1}{2}\exp\left(-e^{\frac{\beta}{2}}\left(t - \frac{n}{2}\right)\right).$$

The rest of the proof then remains essentially unchanged.

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